



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数精讲

开课单位: 数学系

考试时长: 110 分钟

命题教师: 线性代数精讲教师团队

题号	1	2	3	4	5	6	7
分值	30 分	10 分	10 分	10 分	10 分	20 分	20 分

本试卷共 (7) 大题, 满分 (110) 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

This test includes 7 questions. Write **all your answers** on the examination book.

Please put away all books, calculators, cell phones and other devices. Please write carefully and clearly in complete sentences. Your explanations are your only representative when your work is being graded.

Unless otherwise noted, vector spaces are over $\mathbb{F}$ and with finite dimensions, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. $M_{n \times n}(\mathbb{F})$ is the vector space consisting of all $n \times n$ matrices with entries from $\mathbb{F}$.

1. (30 points, 6 points each) Short Questions (Along with your answer, provide an informal proof, counterexample, or other explanation).

- (a) For a fixed $a \in \mathbb{R}$, determine the dimension of the subspace of $\mathcal{P}_n(\mathbb{R})$ defined by $\{f \in \mathcal{P}_n(\mathbb{R}) : f(a) = 0\}$.
- (b) Find a polynomial $p(x) \in \mathcal{P}_2(\mathbb{R})$, such that $p(1) = 8$, $p(2) = 5$, $p(3) = -4$.
- (c) Define the linear map $T : \mathcal{P}_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ by

$$T(f(x)) = \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix}.$$

Find $\dim \text{range } T$.

- (d) If $T : V \rightarrow W$ is a linear map and $v_1, v_2, \dots, v_n$ are vectors in V such that $Tv_1, Tv_2, \dots, Tv_n$ are linearly independent. Show that $v_1, v_2, \dots, v_n$ are linearly independent.
- (e) Let U be subspace of V and let $T : U \rightarrow W$ be a nonzero linear map. Suppose that the function

$$S(v) := \begin{cases} Tv, & \text{if } v \in U; \\ 0, & \text{if } v \notin U. \end{cases}$$

is a linear map. Show that $U = V$.

2. (10 points) Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.
3. (10 points) Let V be a finite dimensional inner product space. Prove that if $u, v \in V$, then

$$|||u|| - ||v||| \leq ||u - v||$$

4. (10 points) Let U, V, W be finite dimensional vector spaces, and $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$. Show that

$$\dim \text{range } T = \dim \text{range } ST + \dim(\text{range } T \cap \text{null } S).$$

5. (10 points) Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

6. (20 points) Let T be the linear operator on $M_{n \times n}(\mathbb{R})$ defined by $T(A) = A^T$.

- (a) Show that 1 and -1 are the only eigenvalues of T .
- (b) Describe the eigenvectors corresponding to each eigenvalue of T .
- (c) Find an ordered basis for $M_{2 \times 2}(\mathbb{R})$ with respect to which T has a diagonal matrix.
- (d) Find an ordered basis for $M_{n \times n}(\mathbb{R})$ ($n > 2$) with respect to which T has a diagonal matrix.

7. (20 points) Let V be a finite dimensional vector space. For a vector $v \in V$, we define $\hat{v} : V' \rightarrow \mathbb{F}$ by $\hat{v}(f) = f(v)$ for every $f \in V'$.

- (a) Show that $\hat{v}$ is a linear functional on V' , i.e. $\hat{v} \in V'' = (V')'$.
- (b) If $\hat{v}(f) = 0$ for all $f \in V'$, then $v = 0$.
- (c) Define $\varphi : V \rightarrow V''$ by $\varphi(v) = \hat{v}$. Show that φ is an isomorphism.